Sleepy Turret

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Problem

Tomorrow, February 28

ABSTRACT ALGEBRA - FOGEL

ABSTRACT ALGEBRA - FOGEL ASSIGNMENT
HW20000: A lot of mathy math
20 PTS
DUE: 8:55 AM

FRENCH IV - KWIATKOWSKI

FRENCH IV - KWIATKOWSKI ASSIGNMENT
200 pg Essay in German Due Yesterday
50 PTS
DUE: 11:59 PM

FRENCH IV - KWIATKOWSKI ASSIGNMENT
Oui Oui Ur Mom
100 PTS
DUE: 3:30 PM

Idea

Hardware

Tower Pro
SG51R
Micro servo
Tower Pro
SG51R
Micro servo

Adobe Stock | #233380090

How do it work?

- Python script detects face, checks if you are sleeping
  - What I'll go over
- Sends coords of camera to Arduino serial, resetting servo position
- Checks boolean if sleeping detected by algorithm
  - Shoots water for 1 second at 5L/min
- Need to calibrate because right now it shoots water while servo is turning

EXPLORER

OPEN EDITORS 1 unsaved

LASER TURRET

- .vscode
- LaserTurretArduino
  - SleepyShooterArduino.i...
- Schematic.png
- SleepyShooter.py

SleepyShooter.py ● | Schematic.png

SleepyShooter.py > ...

```python
import cv2
import serial,time
face_cascade = cv2.CascadeClassifier(cv2.data.haarcascades + 'haarcascade_frontalface_default.xml'

cap=cv2.VideoCapture(0)

ArduinoSerial=serial.Serial('com3',9600,timeout=0.1)
time.sleep(1)

while cap.isOpened():
    ret, frame= cap.read()
    frame=cv2.flip(frame,1)  #mirror the image
    #print(frame.shape)
    gray = cv2.cvtColor(frame,cv2.COLOR_BGR2GRAY)
    faces= face_cascade.detectMultiScale(gray,1.1,6)  #face detect
    for x,y,w,h in faces:
        #sending coords to Arduino Serial
        string='X{0:d}Y{1:d}'.format((x+w//2),(y+h//2))
        print(string)
        ArduinoSerial.write(string.encode('utf-8'))
        #center of ur face
        cv2.circle(frame,(x+w//2,y+h//2),2,(0,255,0),2)
        #plot the roi
        cv2.rectangle(frame,(x,y),(x+w,y+h),(0,255,0),3)
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL

Filter (e.g. text, !exclude)

File Edit Sketch Tools Help

LaserTurretArduino

```arduino
#include<Servo.h>

Servo x, y;
int width = 640, height = 480;  // total resolution of the video
int xpos = 70, ypos = 179;  // initial positions of both Servos
int laserPin = 11;
int buzzerPin = 8;
void setup() {

  Serial.begin(9600);
  x.attach(10);
  y.attach(9);
  x.write(xpos);
  y.write(ypos);
  pinMode(laserPin, OUTPUT);
  pinMode(buzzerPin, OUTPUT);


}
const int angle = 2.5;   // degree of increment or decrement

void loop() {
  if (Serial.available() > 0)
  {
    int x_mid, y_mid;
    if (Serial.read() == 'X')
```

Done uploading.

```
Sketch uses 5014 bytes (15%) of program storage space. Maximum is 32256 bytes.
Global variables use 251 bytes (12%) of dynamic memory, leaving 1797 bytes for loca
```

File Edit Sketch Tools Help

relay

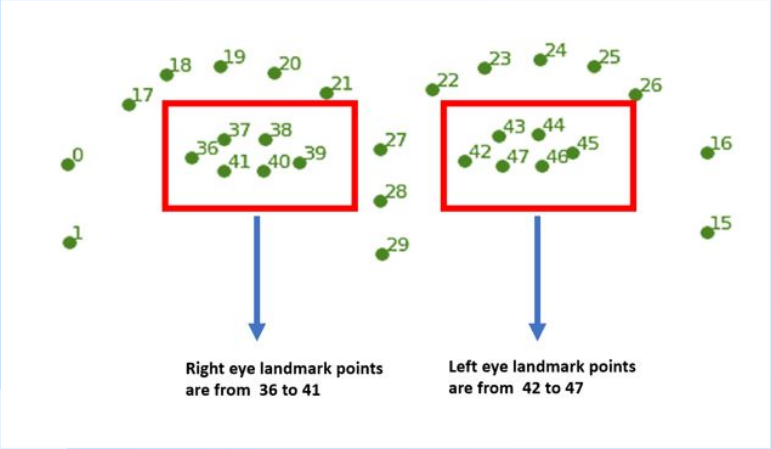
```arduino
int relayPin = 2;
void setup() {
  // put your setup code here, to run once:
  pinMode(relayPin, OUTPUT);
}

void loop() {
  // put your main code here, to run repeatedly:
  digitalWrite(relayPin, HIGH);
  delay(2000);
  digitalWrite(relayPin, LOW);
  delay(2000);
}
```

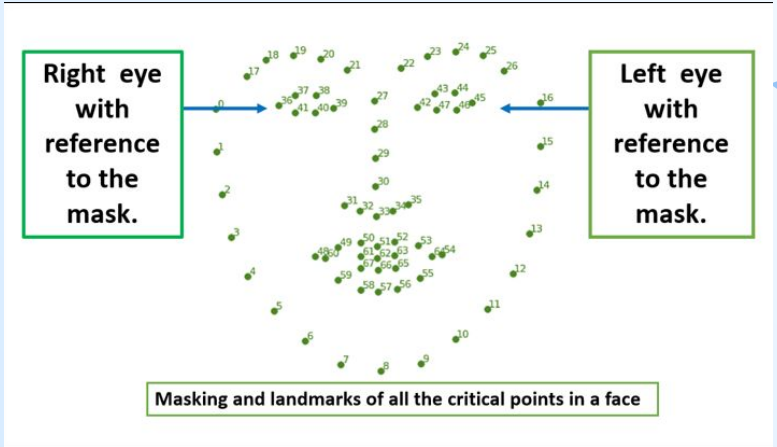
Done uploading.

```
Sketch uses 936 bytes (2%) of program storage space. Maximum is 32256 bytes.
Global variables use 9 bytes (0%) of dynamic memory, leaving 2039 bytes for local
```

How do it work?

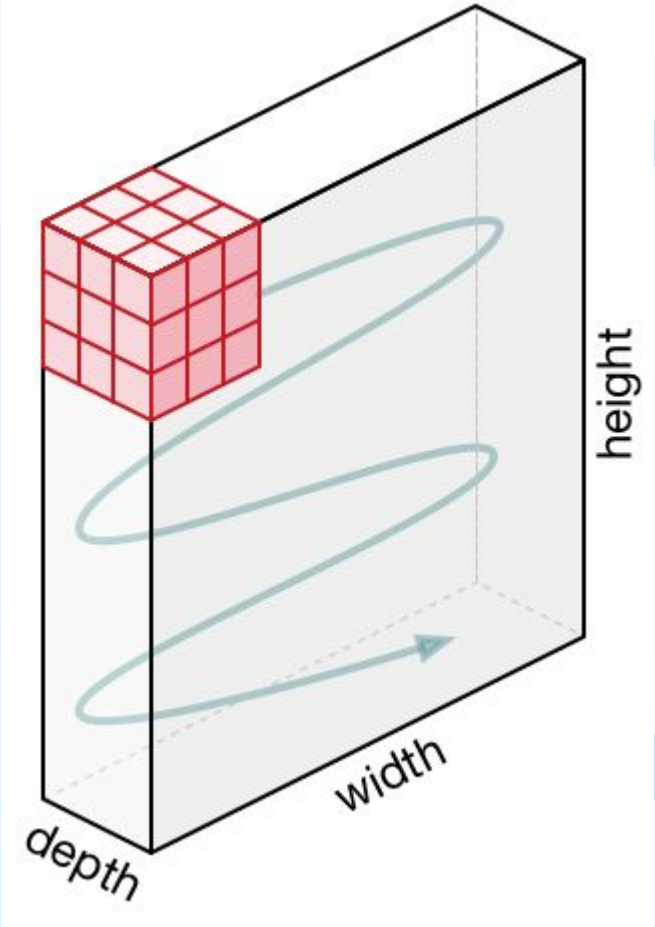
1. Takes input from camera
2. Detect Face in the Image and Create a Region of Interest (ROI)
   a. Converts image to grayscale, as we do not require color to detect faces
3. Sends ROI of left and right eyes using OpenCV FaceMap feature and sends to CNN classifier (will explain next slide)
4. Normalize data of pixels with r_eye = r_eye/255 (all values will be from 0-1
5. lpred = model.predict_classes(l_eye) and same for r_eye, will send true if closed
6. >5 seconds => shoot water





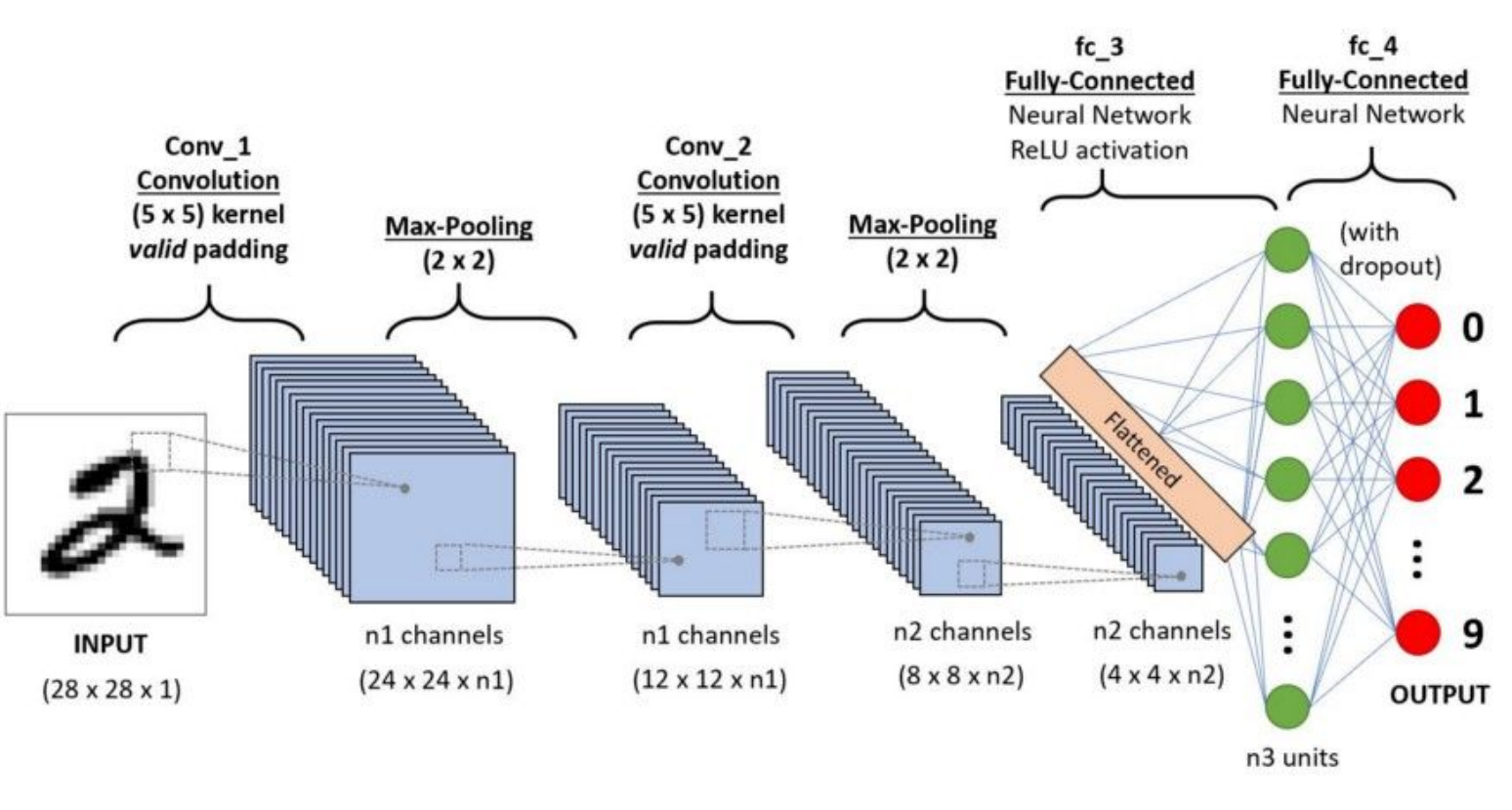
How does a CNN work?

K =		
1	0	1
0	1	0
1	0	1

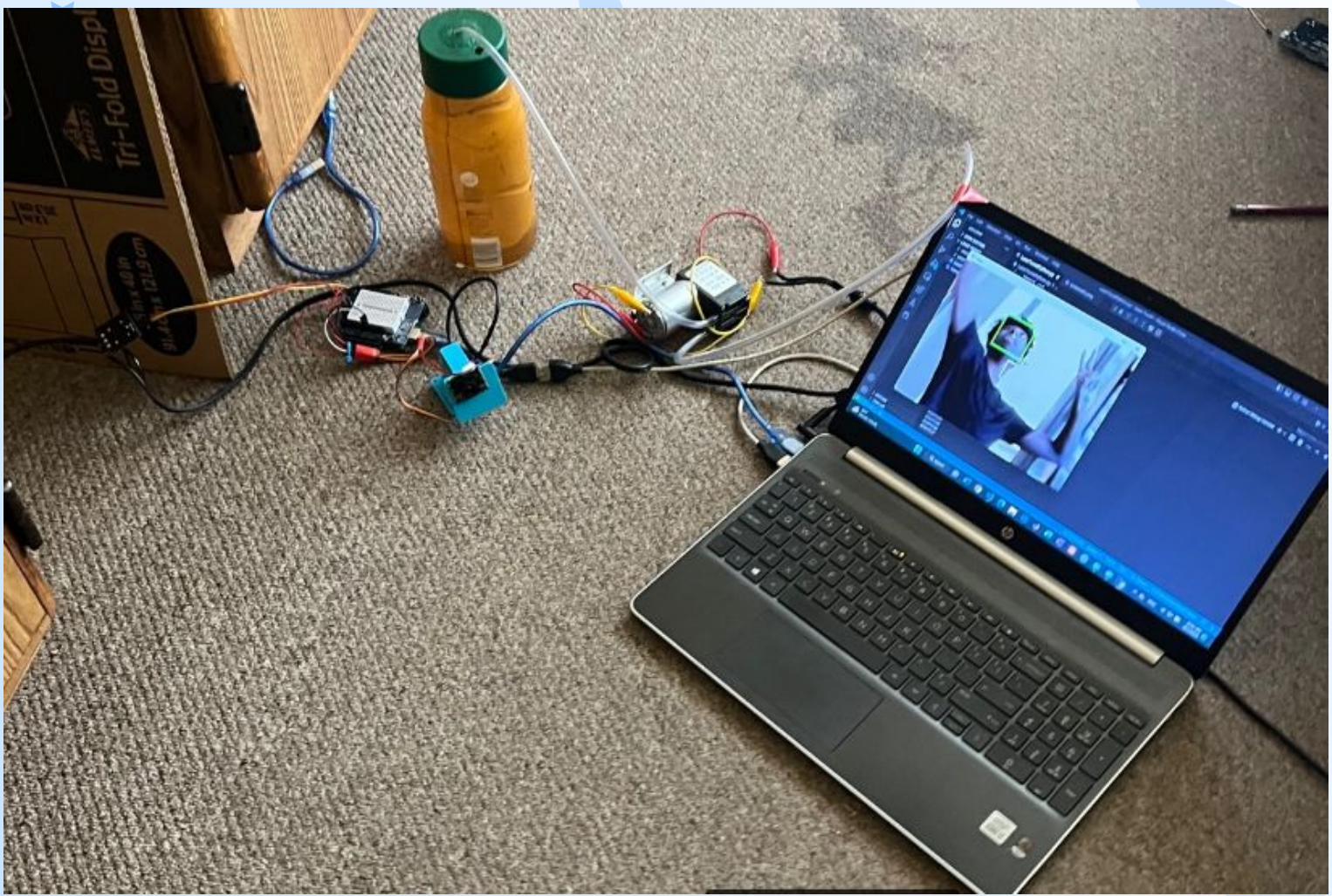


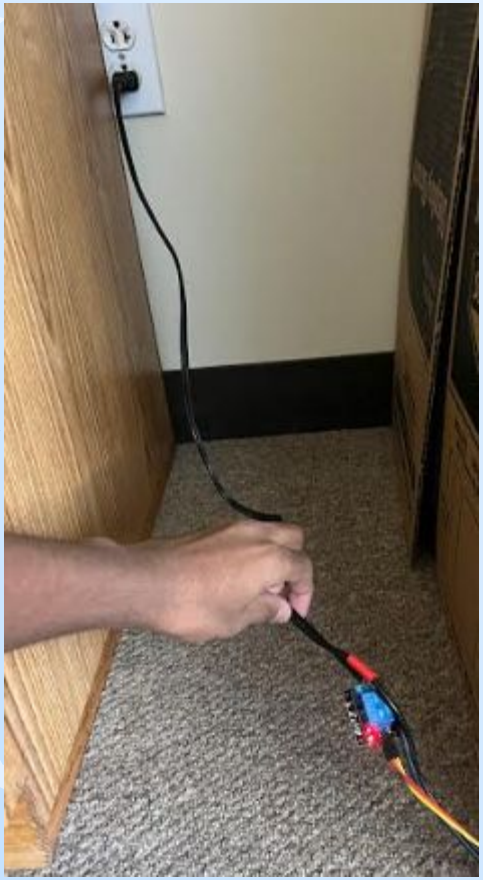
1. Detects high-level features of image
2. Includes edges of the image too
3. Not limited to high-level features, and can perform on color and gradient orientation

Diagram

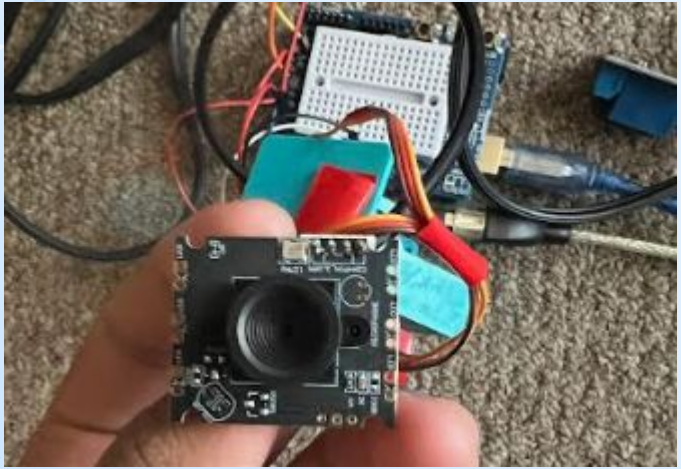

Conv_1
Convolution
(5 x 5) kernel
valid padding
Max-Pooling
(2 x 2)
Conv_2
Convolution
(5 x 5) kernel
valid padding
Max-Pooling
(2 x 2)
fc_3
Fully-Connected
Neural Network
ReLU activation
fc_4
Fully-Connected
Neural Network
(with
dropout)
Flattened
INPUT
(28 x 28 x 1)
n1 channels
(24 x 24 x n1)
n1 channels
(12 x 12 x n1)
n2 channels
(8 x 8 x n2)
n2 channels
(4 x 4 x n2)
n3 units
0
1
2
9
OUTPUT

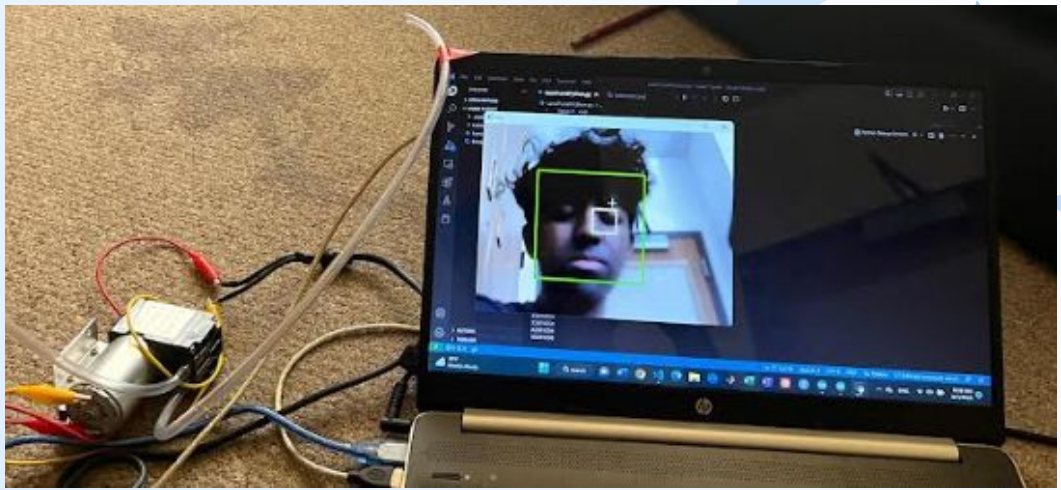
Finished Product?

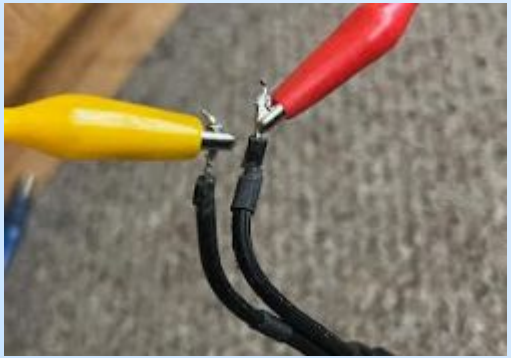

Tri-Fold Displ

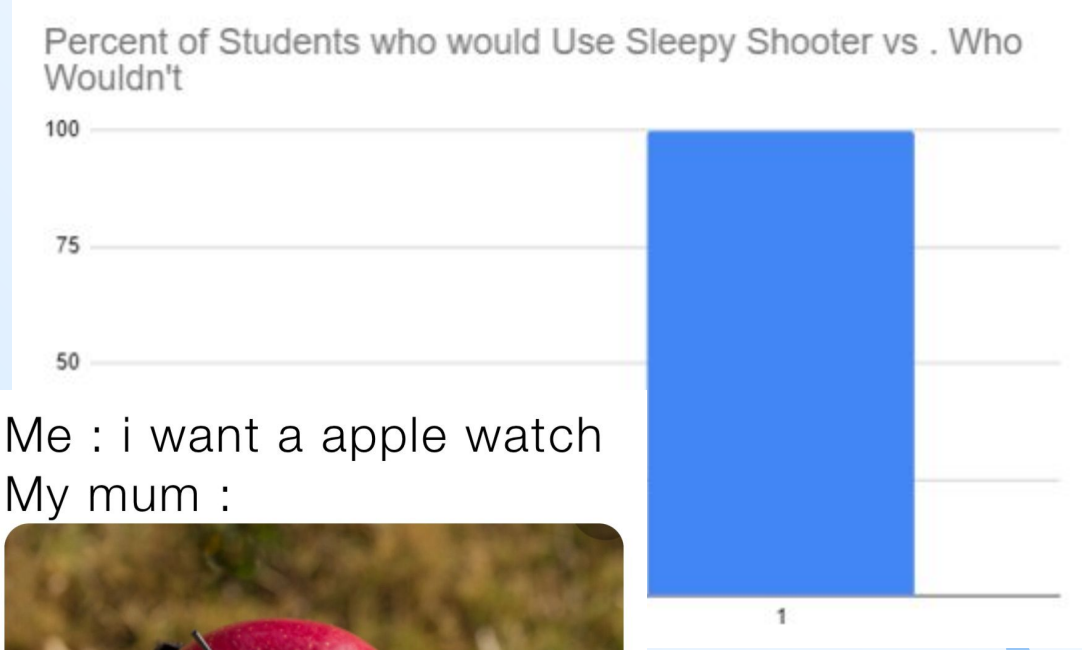


ICED ESPRESSO







Usefulness

- IMSA students would find this useful
- Less Driving accidents by sleepy ppl
- Tesla incorporates sleep detection using cabin based cameras
- Every year. driving accounts for about 100,000 crashes, 71,000 injuries and 1,550 fatalities
- Obv. in seriousness can detect sleep with other sensors like heartbeat, sound, etc.
- Think apple watch stuff